Q10. Sleuth Kit partition and filesystem examination

Original question: Examine a supplied forensic disk image using The Sleuth Kit. Identify the partition layout, file system type, and relevant file system information wherever available.

Aim

Identify the supplied image's partition layout, filesystem type and available filesystem metadata.

Requirements

The Sleuth Kit, a verified image/segments, known sector size when available, and read access. No filesystem mounting is needed. References: [R8], [R9].

Procedure

Step 1 - Identify the image and volume system

```
mkdir -p "$HOME/EHF_Exam/Q10"
cd "$HOME/EHF_Exam/Q10"
IMAGE='<DISK_IMAGE_PATH>'
sha256sum "$IMAGE" | tee image_before.sha256
img_stat "$IMAGE" | tee image_info.txt
mmls "$IMAGE" | tee partitions.txt
```

`img_stat` reports image information supported by the installed build. `mmls` shows partition-table entries, starts, ends and lengths in sectors. Read its unit size; do not treat its start sector as a byte offset. File extensions alone do not establish the image format.

Step 2 - Select a real filesystem partition

```
START='<PARTITION_START_SECTOR>'
SECTOR_BYTES='<DEVICE_SECTOR_SIZE_BYTES>'
fsstat -b "$SECTOR_BYTES" -o "$START" "$IMAGE" > filesystem.txt
sed -n '1,120p' filesystem.txt
```

Replace START with the Start column for the selected data partition, not the slot number or an unallocated/partition-table entry. Set SECTOR_BYTES from image metadata/acquisition facts (commonly 512, sometimes 4096). `-o` is a sector count; `-b` specifies device sector size. fsstat's block/cluster size is a different quantity. For each relevant partition repeat with its own offset and distinct output filename.

Step 3 - Handle image variants and verify integrity

For an image containing only a filesystem/partition, mmls can legitimately find no partition table. Set START=0 and run fsstat if the evidence format supports that interpretation. A file-level text copy from Q6 has neither a partition table nor filesystem and will not work. For supported E01/split images, preserve all segments and let the supported image library read them; use `img_stat -i list` to check formats. Do not force raw mode onto a compressed container. Encrypted/unsupported filesystems need the lab-approved tooling.

```
sha256sum -c image_before.sha256
```

For segmented input, this check covers only the file originally hashed; repeat for every segment or use a manifest containing all segments.

Expected output - illustrative interpretation

```
mmls units: 512-byte sectors
Selected data partition start: 2048
fsstat filesystem type: NTFS
```

This is a teaching summary, not a fabricated partition table. For these illustrative numbers, byte offset = 2048 x 512 = 1,048,576 bytes. Pass `-o 2048 -b 512`, not `-o 1048576`. Your filesystem could instead be FAT, ext or another supported type.

Interpretation, observation and result

Record partition slot, start/end/length, sector size and description; confirm the filesystem using fsstat rather than trusting a partition type label. Record volume label/serial when present, sector and cluster/block sizes, metadata range and other relevant facts available. Unallocated disk regions lie outside listed data partitions; filesystem free space lies inside a filesystem. If no layout is available, report the supported reason and do not invent entries. The result is a documented layout and filesystem analysis for the selected volumes.

Common errors

- Cannot determine filesystem: verify offset, sector size, image format and partition choice before forcing a type.
- Missing image segment: restore the complete ordered set and verify hashes.
- mmls fails on a partition image: try the documented filesystem-at-zero branch, not an arbitrary offset.

Viva questions

- What does mmls examine? A volume system/partition layout.
- What does fsstat examine? Details of a filesystem at the selected offset.
- What are the units of -o? Device sectors, not bytes.
- Is cluster size the same as sector size? No; a cluster can contain multiple sectors.